DATE: / /
PAGE NO.:

Assignment - 1

Q1 Define following matrix
a) orthogonal Matrix
b) Skew symmetric Matrix
c) Symmetric Matrix

Ans (a) The product of a square Matrix and its transpose gives an identity matrix then the square matrix is known as orthogonal matrix

(b) Skew symmetric matrix is defined as the square matrix that is equal to negative of its transpose matrix

(c) Symmetric matrix is defined as a matrix that is equal to its transpose.

Q2 Obtain the characterstics equation of matrix and verify that it is satisfy by A and hence find its inverse

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

Sol^n $$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 2 \\ 0 & 2-\lambda & 1 \\ 2 & 0 & 3-\lambda \end{vmatrix}$$

$$= (1-\lambda)\left[(2-\lambda)(3-\lambda)\right] - 0 + 2(-4+2\lambda) = 0$$

$$= \lambda^3 - 6\lambda^2 + 7\lambda + 2 = 0 \leftarrow$$ Characterstics equation

DATE: / /

PAGE NO.:

Satisfy by putting $\lambda = A$.

$$A^3 - 6A^2 + 7A + 2 = 0$$

$$A^2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix}$$

$$A^3 - 6A^2 + 7A + 2I$$

$$\begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 33 \\ 34 & 0 & 55 \end{bmatrix} - \begin{bmatrix} 30 & 0 & 48 \\ 12 & 24 & 30 \\ 48 & 0 & 78 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 14 \\ 0 & 14 & 7 \\ 14 & 0 & 21 \end{bmatrix}$$

$$+ \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = 0 \quad \text{HP.}$$

for A^{-1}

$$(A^3 - 6A^2 + 7A + 2)A^{-1} = 0 \times A^{-1}$$

$$A^2 - 6A + 7I + 2A^{-1} = 0$$

$$-2A^{-1} = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} - \begin{bmatrix} 6 & 0 & 12 \\ 0 & 12 & 6 \\ 12 & 0 & 18 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -3 & 0 & 2 \\ -1 & 1/2 & 1/2 \\ 2 & 0 & -1 \end{bmatrix}$$

Ans

DATE: / /

PAGE NO.:

Wrong Ques

Q3 Given that $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$. compute the value of $A^6 - 5A^5 + 8A^4 - 2A^3 - 9A^2 + 31A - 36I$ using cayley Hamilton Theorem.

Solⁿ $|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 3 \\ 2 & 1-\lambda & -1 \\ 1 & -1 & 1-\lambda \end{vmatrix}$

$= (1-\lambda)\left[(1-\lambda)^2 - 1\right] + 3\left[-2 - 1 + \lambda\right]$

$= -\lambda^3 + 3\lambda^2 + \lambda - 9$

$= A^3 - 3A^2 - A + 9 = 0$

$A^2 = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -3 & 6 \\ 3 & 2 & 4 \\ 0 & 0 & 5 \end{bmatrix}$

$A^3 = \begin{bmatrix} 4 & -3 & 6 \\ 3 & 2 & 4 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -9 & 21 \\ 11 & -2 & 11 \\ 5 & -5 & 5 \end{bmatrix}$

$A^3 - 3A^2 - A + 9$

$\begin{bmatrix} 4 & -9 & 21 \\ 11 & -2 & 11 \\ 5 & -5 & 5 \end{bmatrix}$

DATE: / /

PAGE NO.:

Q4 Find the eigen values and eigen vector of the matrix

$$\begin{bmatrix} -2 & 1 & 1 \\ -11 & 4 & 5 \\ -1 & 1 & 0 \end{bmatrix}$$

sol^n $|A - \lambda I| = \begin{vmatrix} -2-\lambda & 1 & 1 \\ -11 & 4-\lambda & 5 \\ -1 & 1 & -\lambda \end{vmatrix}$

$= (-2-\lambda)\left[-4\lambda + \lambda^2 - 5\right] - 1\left[11\lambda + 5\right] + 1\left[-11 + 4 - \lambda\right]$

$= 8\lambda - 2\lambda^2 + 10 + 4\lambda^2 - \lambda^3 + 5\lambda - 11\lambda - 5 - 7 - \lambda$

$= -\lambda^3 + 2\lambda^2 + \lambda - 2 = 0$

$\lambda = 1$ is factor

$$\begin{array}{r|l} \lambda - 1 & \lambda^3 - 2\lambda^2 - \lambda + 2 \quad (\lambda^2 - \lambda - 2 \\ & -\lambda^3 \mp \lambda^2 \\ & -\lambda^2 - \lambda \\ & -\lambda^2 + \lambda \\ & + \quad - \\ & -2\lambda + 2 \\ & -2\lambda + 2 \\ & 0 \end{array}$$

$\lambda^2 - \lambda - 2 = 0$

$(\lambda + 1)(\lambda - 2) = 0$

$\lambda = -1, 2, 1 \leftarrow$ eigen values.

For eigen vector

- $\lambda = 1$

$$\begin{bmatrix} -3 & 1 & 1 \\ -11 & 3 & 5 \\ -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$= -3x_1 + x_2 + x_3 = 0$

$-11x_1 + 3x_2 + 5x_3 = 0$

$-x_1 + x_2 - x_3 = 0$

DATE: / /

PAGE NO.:

by using crammers Rule

$$\frac{x_1}{\begin{vmatrix} 1 & 1 \\ 3 & 5 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} -3 & 1 \\ -11 & 5 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} -3 & 1 \\ -11 & 3 \end{vmatrix}}$$

$$\frac{x_1}{2} = \frac{x_2}{4} = \frac{x_3}{2} = K \Rightarrow \quad x_1 = 2 \quad x_3 = 1$$

$$x_2 = 2$$

$$\text{eigen vector} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$\lambda = -1$

$$= \begin{bmatrix} -1 & 1 & 1 \\ -11 & 5 & 5 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_4 \\ x_5 \\ x_6 \end{bmatrix} \Rightarrow \begin{aligned} -x_4 + x_5 + x_6 &= 0 \\ -11x_4 + 5x_5 + 5x_6 &= 0 \\ -x_4 + x_5 + x_6 &= 0 \end{aligned}$$

by using crammer's Rule

$$\frac{x_4}{\begin{vmatrix} 1 & 1 \\ 5 & 5 \end{vmatrix}} = \frac{-x_5}{\begin{vmatrix} -1 & 1 \\ -11 & 5 \end{vmatrix}} = \frac{x_6}{\begin{vmatrix} -1 & 1 \\ -11 & 5 \end{vmatrix}}$$

$$= \frac{x_4}{0} = \frac{-x_5}{6} = \frac{x_6}{6} \qquad \text{eigen vector} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$x_4 = 0, \; x_5 = -1, \; x_6 = 1$

$\lambda = 2$

$$= \begin{bmatrix} 0 & 1 & 1 \\ -11 & 2 & 5 \\ -1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_7 \\ x_8 \\ x_9 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 0x_7 + x_8 + x_9 &= 0 \\ -11x_7 + 2x_8 + 5x_9 &= 0 \\ -x_7 + x_8 - 2x_9 &= 0 \end{aligned}$$

DATE: / /
PAGE NO.:

By using cramer's Rule

$$\frac{x_7}{\begin{vmatrix} 1 & 1 \\ 12 & 5 \end{vmatrix}} = \frac{-x_8}{\begin{vmatrix} 0 & 1 \\ -11 & 5 \end{vmatrix}} = \frac{x_9}{\begin{vmatrix} 0 & 1 \\ -11 & 2 \end{vmatrix}}$$

$$= \frac{x_7}{3} = \frac{-x_8}{11} = \frac{x_9}{11} = K$$

$x_7 = 3$ $\quad x_8 = -11$ $\quad x_9 = 11$

eigen vector $= \begin{bmatrix} 3 \\ -11 \\ 11 \end{bmatrix}$

Assignment - 2

Q1 Find the inverse of Matrix with the help of cayley Hamilton Theorem.

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Sol^n $|A - \lambda I| = \begin{bmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ +1 & -1 & 2-\lambda \end{bmatrix}$

$= (2-\lambda)[(2-\lambda)(2-\lambda) - 1] + 1\cancel{[1 + 2 - \lambda]}[-2 + \lambda + 1]$

$+ 1[1 - 2 + \lambda]$

$= (2-\lambda)[4 - 4\lambda + \lambda^2 - 1] + \lambda - 1 + \lambda - 1$

$= 8 - 8\lambda + 2\lambda^2 - 2 - 4\lambda + 4\lambda^2 - \lambda^3 + \lambda + \lambda - 1 + \lambda - 1$

$= -\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0$

DATE: / /

PAGE NO.:

In cayley Hamilton theorem put $\lambda = A$

$$(A^3 - 6A^2 + 9A - 4I = 0)\ A^{-1}$$

$$A^2 - 6A + 9I - 4A^{-1} = 0$$

$$4A^{-1} = A^2 - 6A + 9I$$

$$A^2 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - \begin{bmatrix} 12 & -6 & 6 \\ -6 & 12 & -6 \\ 6 & -6 & 12 \end{bmatrix} + \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = 4A^{-1}$$

$$= \frac{1}{4} \begin{bmatrix} 27 & -11 & 11 \\ -11 & 27 & -11 \\ 11 & -11 & 27 \end{bmatrix} = A^{-1}$$ ans

Q2 Find the eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

sol^n $$|A - \lambda I| = \begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{vmatrix}$$

$$= (-2-\lambda)\left[(1-\lambda)(-\lambda) - 12\right] - 2\left[-2\lambda - 6\right] - 3\left[-4 + 1 - \lambda\right] = 0$$

$$\Rightarrow (-2-\lambda)\left[-\lambda + \lambda^2 - 12\right] + 4\lambda + 12 + 9 + 3\lambda = 0$$

$$\Rightarrow +2\lambda - 2\lambda^2 + 24 + \lambda^2 - \lambda^3 + 12\lambda + 7\lambda + 21 = 0$$

$$= -\lambda^3 - \lambda^2 + 21\lambda + 45 = 0$$

$$\Rightarrow \lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

$$\Rightarrow (\lambda + 3)^2 (\lambda - 5) = 0$$

$$\lambda = -3, 5$$

DATE: / /

PAGE NO.:

eigen vector for $\lambda = -3$

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow x_1 + 2x_2 - 3x_3 = 0 \quad -(1)$$
$$2x_1 + 4x_2 - 6x_3 = 0 \quad -(2)$$
$$-x_1 - 2x_2 + 3x_3 = 0 \quad -(3)$$

Crammer's Rule in eq (1) & (2)

$$\frac{x_1}{\begin{vmatrix} 2 & -3 \\ 4 & -6 \end{vmatrix}} = \frac{x_2}{\begin{vmatrix} 1 & -3 \\ 2 & -6 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}}$$

$$= \frac{x_1}{0} = \frac{x_2}{0} = \frac{x_3}{0}$$

eigen vector : $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

eigen vector for $\lambda = 5$

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_4 \\ x_5 \\ x_6 \end{bmatrix} = 0$$

$$\Rightarrow -7x_4 + 2x_5 - 3x_6 = 0 \quad -(1)$$
$$2x_4 - 4x_5 - 6x_6 = 0 \quad -(2)$$

$$\Rightarrow \frac{x_4}{\begin{vmatrix} 2 & -3 \\ -4 & -6 \end{vmatrix}} = \frac{x_5}{\begin{vmatrix} -7 & -3 \\ 2 & -6 \end{vmatrix}} = \frac{x_6}{\begin{vmatrix} -7 & 2 \\ 2 & -4 \end{vmatrix}}$$

$$= \frac{x_4}{-24} = \frac{x_5}{48} = \frac{x_6}{24}$$

eigen vector $= \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ ans

DATE: / /
PAGE NO.:

Q3 Reduce the Matrix $A = \begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$ to digonal form.

Sol^n $|A - \lambda I| = \begin{vmatrix} -1-\lambda & 2 & -2 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & -\lambda \end{vmatrix}$

$= (-1-\lambda)[-2\lambda + \lambda^2 + 1] - 2[-\lambda + 1] - 2[-1 + 2 - \lambda] = 0$

$= 2\lambda - \lambda^2 - 1 + 2\lambda^2 - \lambda^3 - \lambda + 2\lambda - 2 - 2 + 2\lambda = 0$

$= -\lambda^3 + \lambda^2 + 5\lambda - 5 = 0$

$\Rightarrow \lambda^3 - \lambda^2 - 5\lambda + 5 = 0$

$\lambda = 1$ is a root

$$\begin{array}{r|l} \lambda - 1 & \lambda^3 - \lambda^2 - 5\lambda + 5 \quad (\lambda^2 - 5 \\ & \underline{\lambda^3 - \lambda^2} \\ & \quad - \quad + \\ & \quad -5\lambda + 5 \\ & \quad \underline{-5\lambda - 5} \\ & \quad 0 \end{array}$$

$\lambda^2 - 5 = 0$

$\lambda = \pm\sqrt{5}$

eigen value $\rightarrow 1, \sqrt{5}, -\sqrt{5}$

eigen vector for $\lambda = 1$

$$\begin{bmatrix} -2 & 2 & -2 \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$-2x_1 + 2x_2 - 2x_3 = 0$

$x_1 + x_2 + x_3 = 0$

$x_1 + x_2 + x_3 = 0$

$x_1 = x_2 = x_3 = 0$

$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

DATE : / /

PAGE NO. :

eigen vector for $\lambda = \sqrt{5}$

$$\begin{bmatrix} -1-\sqrt{5} & 2 & -2 \\ 1 & 2-\sqrt{5} & 1 \\ -1 & -1 & -\sqrt{5} \end{bmatrix} \begin{bmatrix} x_4 \\ x_5 \\ x_6 \end{bmatrix} = 0$$

$$\Rightarrow (-1 - \sqrt{5})x_4 + 2x_5 - 2x_6 = 0$$

$$x_4 + (2-\sqrt{5})x_5 + x_6 = 0$$

$$-x_4 - x_5 - \sqrt{5}x_6 = 0$$

$$= \frac{x_4}{\begin{vmatrix} 2-\sqrt{5} & 1 \\ -1 & -\sqrt{5} \end{vmatrix}} =$$